Boundary partitions at the leximin core: a two-parameter characterization

Maxwell Flanagan

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Abstract

At a slack strict payoff boundary of the leximin core allocation of a singleton-normalized exact game, the two richer players may each carry a tight coalition, and those two coalitions may partition the player set. We characterize exactly when: writing p and q for the number of poorer players on the two sides, such a *boundary partition* exists if and only if $p, q \geq 1$ and $p + q \geq 4$. Both directions are proved in closed form. The obstruction and the construction turn out to be the same inequality read in opposite directions: a tight coalition that contains the boundary and omits one side's poorer players costs two spare players on the other side. The characterization settles a residual question about zero-payoff configurations on seven players, in the affirmative rather than, as we had conjectured, the negative. Two methodological ingredients are of independent use: exactness of a game is an existential linear condition, needing no semi-balanced catalogue and no vertex enumeration; and, at a fixed payoff stratification, leximin optimality is a Farkas certificate supported on the tight coalitions, hence quantifier-free.

1 Introduction

Let v be a transferable-utility game on a finite player set N that is *exact*, meaning $v(S) = \min_{x \in \mathrm{Core}(v)} x(S)$ for every coalition, and *singleton-normalized*, meaning $v(\{i\}) = 0$ for every player. Let $\phi = \mathrm{Lmin}(v)$ be its leximin core allocation, and let $N = L \,\dot\cup\, H$ be a strict payoff boundary: H is a union of the richest payoff strata of ϕ and $\max_{i \in L} \phi_i < \min_{j \in H} \phi_j$. The boundary is *slack* when $v(H) < \phi(H)$. A companion paper (Flanagan, 2026) determines exactly which pairs $(|L|, |H|)$ admit a slack boundary.

This note answers a finer question at the smallest interesting richer side, $|H| = 2$. Write $H = \{f, g\}$.

Definition 1 (boundary partition)**.** *A* boundary partition of type (p, q) *at a slack strict boundary* $L \,\dot\cup\, H$ *with* $H = \{f, g\}$ *is a pair of* ϕ*-tight coalitions*

$$S_1 = P \cup \{f\}, \qquad S_2 = Q \cup \{g\}, \qquad P \,\dot\cup\, Q = L, \quad p = |P|, \ q = |Q|,$$

so that S_1 *and* S_2 *are disjoint, cover* N*, and meet* H *in one player each. We write* $\mathrm{BP}(p, q)$ *for the type, and* $m = p + q = |L|$*,* $n = m + 2 = |N|$*.*

The configuration is the extreme case of the failure of exact partition: not only is the richer suffix H not tight, but the two richer players separately anchor a partition of N into tight blocks. Our main result is that this is possible exactly on a two-parameter staircase.

Theorem 2 (characterization)**.** *A boundary partition of type* (p, q) *exists if and only if*

$$p \geq 1, \qquad q \geq 1, \qquad p + q \geq 4.$$

Necessity is Theorem 18; sufficiency is Theorem 22, by two explicit families of games. Neither direction uses a solver.

Three features of the proof are worth advertising.

One inequality, read twice. Everything turns on a single consequence of the uncrossing budget for tight coalitions (Lemma 3): if T is ϕ-tight, contains H, and omits every player of P, then

$$|Q \setminus T| \;\geq\; 2. \tag{1}$$

Read as a prohibition, (1) kills every type with $p + q \leq 3$. Read as a design constraint, it dictates the two realizing families and their ranges. The threshold is therefore not two facts that happen to meet at $m = 4$; it is one inequality that first becomes satisfiable there.

A three-line normal form. Because S_1 and S_2 are complementary tight coalitions, *every* core point is tight on both (Lemma 11). The core therefore collapses to a polytope in $|L|$ variables, and the 2^n coalition functionals collapse onto a short explicit list whose facet normals can never separate two players lying in the same block (Lemma 12). That last sentence is the combinatorial content of the configuration and is what the necessity proof consumes.

A correction. An earlier round of this program conjectured that a boundary partition requires $\max(p, q) \geq 3$, on the strength of a census in which type $\mathrm{BP}(2, 2)$ never appeared among some twenty thousand sampled boundaries. That conjecture is false: $\mathrm{BP}(2, 2)$ is realizable, and Section 9 exhibits a six-player game carrying it. The correct threshold is on $m = p + q$, not on $\max(p, q)$. We record this because the failure mode is a general hazard: absence from a random census is evidence about sampling, not about realizability, and the same caution is raised in (Flanagan, 2026) for a different configuration. The consequence for the seven-player zero-payoff question of Section 9 is that it resolves in the opposite direction from the one we expected.

Prior art. [TODO: prior-art positioning; the lead runs the citation verification pass and no reference below has been invented.] Citations are needed for: the egalitarian core and its transfer-blocking characterization, which is where the protector Lemma 4 sits in the literature; exact games and large cores, for the standing hypotheses of Section 2; Dutta–Ray egalitarianism and Lorenz domination, which is the context in which exact partition at Lmin is normally discussed; and any prior appearance of an existential characterization of exactness in the spirit of Theorem 8, which we believe to be folklore but have not traced. The companion paper (Flanagan, 2026) carries the corresponding positioning for the phase diagram and is the right place to start.

Sections 2 and 3 fix notation and record the descent that reduces zero-payoff configurations to boundary partitions. Section 4 isolates the two encoding theorems. Section 5 sets up the normal form, Section 6 proves necessity, Section 7 gives the families. Section 8 describes what happens exactly at the threshold, and Section 9 draws the seven-player consequence. Sections 10 and 11 state the status of every claim and the limits of the result.

2 Preliminaries

All games are on a finite set N, exact, and singleton-normalized; core allocations are then nonnegative. For $\phi \in \mathrm{Core}(v)$ write $s_\phi(S) = \phi(S) - v(S)$ for the slack, and call S *ϕ-tight* when $s_\phi(S) = 0$. We use $\mathrm{Lmin}(v)$ for the leximin-maximal core allocation, computed in the usual stages: at each

stage one maximizes the common lower bound on the payoffs of the not-yet-pinned players subject to the pinned ones being fixed, and pins those players whose payoff cannot exceed that bound. The payoff strata of ϕ are its level sets, ordered $B_1 \prec \cdots \prec B_k$ from poorest to richest, and a *richer suffix* is a union $B_{j+1} \cup \cdots \cup B_k$.

We use two facts from the companion paper. The first is the uncrossing budget.

Lemma 3 (crossing budget)**.** *Let v be exact and singleton-normalized and $\phi \in \mathrm{Core}(v)$. If S_1 and S_2 are ϕ-tight then*

$$s_\phi(S_1 \cap S_2) \;\le\; \phi\big(N \setminus (S_1 \cup S_2)\big).$$

The second is the existence of protectors, which we restate with a direct proof because it is used repeatedly.

Lemma 4 (protectors)**.** *Let $\phi = \mathrm{Lmin}(v)$. If $\phi_i < \phi_j$ then some proper ϕ-tight coalition contains j and omits i.*

Proof. Otherwise $d = e_i - e_j$ satisfies $d(T) = [i \in T] - [j \in T] \ge 0$ for every ϕ-tight T, the value -1 occurring only when $j \in T$ and $i \notin T$. Also $d(N) = 0$, so $\phi + td \in \mathrm{Core}(v)$ for all small $t > 0$. Choose $0 < t < (\phi_j - \phi_i)/2$. The sorted payoff vector then changes only by replacing ϕ_i with $\phi_i + t$ and ϕ_j with $\phi_j - t$, and $\phi_j - t > \phi_i + t$, so no coordinate falls to or below ϕ_i: the first position at which the two sorted vectors differ is strictly larger for $\phi + td$, contradicting leximin-maximality. □

Finally we use the standard lower-envelope construction.

Lemma 5 (envelope)**.** *Let G be a finite family of nonnegative allocations with a common total T, and set $v(S) = \min_{g \in G} g(S)$. Then v is exact and every $g \in G$ lies in $\mathrm{Core}(v)$; and v is singleton-normalized as soon as every coordinate vanishes in some row of G.*

3 Zero-payoff configurations and the descent

Boundary partitions arose from a neighbouring question. Keep $H = \{f, g\}$ and let S_1, S_2 be ϕ-tight with $S_1 \cap H = \{f\}$ and $S_2 \cap H = \{g\}$. A *zero-payoff configuration* of type $\mathrm{ZP}(p, q)$ has $S_1 \cap S_2 = \{e\}$ for a single $e \in L$ and $S_1 \cup S_2 = N$, with $p = |S_1 \cap L \setminus \{e\}|$ and $q = |S_2 \cap L \setminus \{e\}|$. Boundary partitions are the degenerate case $S_1 \cap S_2 = \emptyset$.

The two are linked by deletion and suspension of a null player. We record the chain; the arguments are short and are given in full in the archived proof file cited in Section 10.

Proposition 6 (descent)**.** *Let v be exact and singleton-normalized with $\phi = \mathrm{Lmin}(v)$ and a slack strict boundary $H = \{f, g\}$.*

- *(i) In a $\mathrm{ZP}(p, q)$ configuration, $\phi_e = 0$.*
- *(ii) If $\min_i \phi_i = 0$ then a single player j has $x_j = 0$ for every $x \in \mathrm{Core}(v)$, and j is null.*
- *(iii) Deleting a null player preserves exactness, normalization, the core, the leximin point, the tightness of every tight coalition, and the slack of every boundary avoiding it.*
- *(iv) Conversely, adjoining a null player to a game with a boundary partition of type (p, q) yields a $\mathrm{ZP}(p, q)$ configuration with one more poorer player.*
- *(v) Hence every $\mathrm{ZP}(p, q)$ configuration descends, one poorer player at a time, either to a boundary partition of type (p, q) or to a zero-payoff configuration of type $(p - 1, q)$ or $(p, q - 1)$.*

Two immediate consequences will be used. First, no boundary partition has $q = 0$: that would make the rich singleton $\{g\}$ tight, forcing $\phi_g = v(\{g\}) = 0$ against strictness. So $p, q \ge 1$, which is the easy half of Theorem 2. Second, by (ii)–(iii) we may assume throughout that no poorer player is null:

Lemma 7 (no null poorer player)**.** *If a boundary partition of type (p, q) exists with $\min_{i\in L} \phi_i = 0$, then one of type $(p-1, q)$ or $(p, q-1)$ exists. Consequently, in a boundary partition with m minimal, $\phi_i > 0$ for every $i \in L$ and no singleton is ϕ-tight.*

4 Two encodings that decide

The results below are proved by hand, but the route to them ran through finite checks, and the two encoding theorems that made those checks decisive are of independent interest. Both replace an apparently intractable condition by an exactly equivalent linear one.

Theorem 8 (exactness is existential)**.** *A game v is exact if and only if for every coalition S there is $z^S \in \mathrm{Core}(v)$ with $z^S(S) = v(S)$.*

Proof. Immediate from $v(S) = \min_{x\in\mathrm{Core}(v)} x(S)$: the minimum is attained exactly when such a z^S exists, and every core point satisfies $x(S) \geq v(S)$. □

The content is what this buys. Exactness is usually imposed either by enumerating core vertices, which is exponential, or through the facet description of the cone of exact games by indecomposable min-semi-balanced systems (Studený and Kratochvíl, 2022; Lohmann et al., 2012), which is available only up to six players and whose sufficiency direction is delicate to validate. Theorem 8 instead adjoins one allocation vector per coalition subject to linear constraints: no catalogue, no enumeration, no completeness hypothesis, and no restriction on n.

The second theorem removes the quantifier in "ϕ is the leximin point".

Theorem 9 (leximin is a Farkas certificate)**.** *Let v be a game, $\phi \in \mathrm{Core}(v)$, and let $B_1 \prec \cdots \prec B_k$ be the payoff strata of ϕ, with common values $\mu_1 < \cdots < \mu_k$. Write $\mathcal{T}$ for the family of proper ϕ-tight coalitions. Then $\phi = \mathrm{Lmin}(v)$ if and only if for every stage t and every $i \in B_t$ there are reals $y_A \geq 0$ (A proper), η, ρ_l for $l \in B_1 \cup \cdots \cup B_{t-1}$, and $\sigma_c \geq 0$ for $c \in B_t$, such that*

$$y_A = 0 \quad \text{for every } A \notin \mathcal{T}, \tag{2}$$

and, for every player j,

$$-\sum_{A\ni j} y_A + \eta + [\,j \in B_{<t}\,]\rho_j - [\,j \in B_t\,]\sigma_j = [\,j = i\,]. \tag{3}$$

Proof. By the staged definition, $\phi = \mathrm{Lmin}(v)$ with these strata if and only if for every t and every $i \in B_t$,

$$\max\{x_i :\ x \in \mathrm{Core}(v),\ x_l = \phi_l\ (l \in B_{<t}),\ x_c \geq \mu_t\ (c \in B_{\geq t})\} = \phi_i. \tag{4}$$

Necessity is the definition. For sufficiency, (4) at stage t forces the stage level to be exactly μ_t — a core point with every unpinned coordinate above μ_t would exceed μ_t at $i \in B_t$ — and forces the pinned set to be exactly B_t, since for i in a later stratum ϕ itself gives $x_i = \phi_i > \mu_t$; induct on t.

Now dualize (4). Its feasible set is a polyhedron containing ϕ, whose constraints are efficiency; $x(A) \geq v(A)$ for proper A, active exactly on $\mathcal{T}$; the pins $x_l = \phi_l$; and $x_c \geq \mu_t$, active exactly for $c \in B_t$, because $\phi_c > \mu_t$ for c in a later stratum. Nonnegativity is the singleton case of $x(A) \geq v(A)$. The tangent cone at ϕ is therefore

$$\mathcal{D}_t = \{d :\ d(N) = 0,\ d(T) \geq 0\ (T \in \mathcal{T}),\ d_l = 0\ (l \in B_{<t}),\ d_c \geq 0\ (c \in B_t)\},$$

and (4) says $d_i \leq 0$ on $\mathcal{D}_t$, i.e. e_i lies in the polar cone. That polar is generated by $-\mathbf{1}_T$ for $T \in \mathcal{T}$, by $-e_c$ for $c \in B_t$, and by the lines through $\mathbf{1}_N$ and through e_l for $l \in B_{<t}$; membership of e_i in it is exactly (3) with (2). Conversely, given such multipliers, for any $d \in \mathcal{D}_t$, $d_i = -\sum_A y_A d(A) + \eta\, d(N) + \sum_l \rho_l d_l - \sum_c \sigma_c d_c \leq 0$, each term being non-positive. □

Remark 10. *The naive dual of* (4) *carries products* $y_A v(A)$*, which are bilinear in the unknowns when the game is itself unknown. Complementary slackness at the optimum* ϕ *removes them:* $y_A > 0$ *forces* A *tight, so* $\sum_A y_A v(A) = \sum_A y_A \phi(A)$*, and pairing* (3) *with* ϕ *reproduces the numerical half of the dual automatically. What survives is* (2)*, a disjunction rather than a product. This is why the condition is decidable in linear arithmetic with Boolean structure.*

5 The block-polytope normal form

Fix a boundary partition of type (p, q) with the notation of Definition 1.

Lemma 11 (block rigidity)**.** *Every* $x \in \mathrm{Core}(v)$ *satisfies* $x(S_1) = v(S_1)$ *and* $x(S_2) = v(S_2)$*. In particular* $v(S_1) + v(S_2) = v(N)$*, and the core lies in a face of codimension one.*

Proof. Tightness at ϕ and efficiency give $v(S_1) + v(S_2) = \phi(S_1) + \phi(S_2) = \phi(N) = v(N)$. For $x \in \mathrm{Core}(v)$, $x(S_1) \geq v(S_1)$ and $x(S_2) \geq v(S_2)$ while $x(S_1) + x(S_2) = x(N) = v(S_1) + v(S_2)$. Two inequalities summing to an equality are equalities. □

Lemma 11 is the precise sense in which a boundary partition is close to a direct sum: the core is confined to the product of the two block faces. It is not a direct sum — the core is in general strictly inside that product — and the gap is exactly what makes the threshold 4 rather than something read off one block at a time.

Write $u_i = x_i - \phi_i$ for $i \in L$ and $U_P = \sum_{i \in P} u_i$, $U_Q = \sum_{i \in Q} u_i$.

Lemma 12 (normal form)**.** *Under Lemma 11 the core is a polytope* $K \subseteq \mathbb{R}^L$ *containing the origin, with* $x_f - \phi_f = -U_P$ *and* $x_g - \phi_g = -U_Q$*. For a coalition* T *put* $\varepsilon_f = [f \in T]$ *and* $\varepsilon_g = [g \in T]$*. Then the displacement functional* $d(T)$ *is the linear form on* $\mathbb{R}^L$ *with coefficients*

$$\nu_i = [i \in T] - \varepsilon_f \ (i \in P), \qquad \nu_i = [i \in T] - \varepsilon_g \ (i \in Q).$$

Consequently, within each block the coefficients lie all in $\{0, 1\}$ *or all in* $\{-1, 0\}$*:* no facet normal separates two players of the same block with opposite signs*. Among proper nonempty coalitions the zero form is attained exactly at* $T = S_1$ *and* $T = S_2$*.*

Proof. Solving $x(S_1) = v(S_1)$ for x_f and $x(S_2) = v(S_2)$ for x_g gives the substitutions. Then $d(T) = \sum_{i \in T \cap L} u_i - \varepsilon_f U_P - \varepsilon_g U_Q$, and collecting the coefficient of u_i gives the display. For $i \in P$ that coefficient is $[i \in T] - \varepsilon_f$, which ranges over $\{0, 1\}$ when $\varepsilon_f = 0$ and over $\{-1, 0\}$ when $\varepsilon_f = 1$; likewise on Q. □

The tangent cone of the core at ϕ is $\mathcal{D} = \{d \in \mathbb{R}^L : \nu_T \cdot d \geq 0 \text{ for every } \phi\text{-tight } T\}$.

Lemma 13 (improving directions)**.** $\mathcal{D} \cap \mathbb{R}^L_{\geq 0} = \{0\}$*. Consequently, for every nonempty* $S \subseteq L$ *there is a* ϕ*-tight* T *with*

$$|T \cap S| \ < \ \varepsilon_f \, |P \cap S| + \varepsilon_g \, |Q \cap S|. \tag{5}$$

Proof. Let $d \in \mathcal{D}$ be nonnegative on L and nonzero. The boundary is strict, so every stratum of L precedes every stratum meeting H; let B_j be the first stratum on which d is nonzero. Since $d \neq 0$ somewhere on L, B_j is a stratum of L, so $d|_{B_j} \geq 0$ and $d|_{B_j} \neq 0$. Then $\phi + td$ lies in the core for small $t > 0$ and raises the sorted payoff vector at the first position where it changes, contradicting $\phi = \mathrm{Lmin}(v)$. For the second assertion, apply the first to $d = \mathbf{1}_S$, which is nonnegative and nonzero, so $\mathbf{1}_S \notin \mathcal{D}$; by Lemma 12, $\nu_T \cdot \mathbf{1}_S = |T \cap S| - \varepsilon_f |P \cap S| - \varepsilon_g |Q \cap S|$ for every T, and some ϕ-tight T makes it negative. □

Remark 14. *We claim only the stated direction. Condition* (5) *for every* S *does* not *in general force* $\mathcal{D} \cap \mathbb{R}^L_{\geq 0} = \{0\}$*: lying outside a convex cone is not preserved under nonnegative combinations, so excluding every indicator* $\mathbf{1}_S$ *from* $\mathcal{D}$ *excludes no other nonnegative vector by itself. Concretely, a normal that is* $+1$ *on part of* P *and* -1 *on part of* Q *can satisfy* $\nu \cdot \mathbf{1}_S < 0$ *while* $\nu \cdot d \geq 0$ *for a nonnegative* d *weighted towards* P*. Everything below uses only the implication as stated; where the cone condition itself is needed — in the constructions of Section 7 — it is imposed directly.*

The case $S = \{i\}$ of (5) recovers Lemma 4 in the form we need: for $i \in P$ there is a tight T with $f \in T$ and $i \notin T$.

Lemma 15 (exactness directions)**.** *There is* $d^* \in \mathcal{D}$ *with* $\sum_{i \in P} d^*_i = \phi_f > 0$*, and symmetrically one with* $\sum_{i \in Q} d_i = \phi_g > 0$*.*

Proof. Exactness at $\{f\}$ gives $z \in \mathrm{Core}(v)$ with $z_f = v(\{f\}) = 0$; Lemma 11 gives $z(P) + z_f = v(S_1) = \phi(P) + \phi_f$, so $\sum_{i \in P}(z_i - \phi_i) = \phi_f$. Put $d^* = z - \phi$, which lies in $\mathcal{D}$ because z is a core point. □

When $p = 1$ this pins the whole of ϕ_f onto a single poorer player. When $p \geq 2$ it says only that some member of P gains, and that slack is exactly what the constructions of Section 7 exploit.

Lemma 16 (crossing exclusion)**.** *Let* T *be* ϕ*-tight.*
(i) If $f \in T$ *then* $s_\phi\big((P \cap T) \cup \{f\}\big) \leq \phi(Q \setminus T) + [g \notin T]\,\phi_g$*.*
(ii) If $g \in T$ *then* $s_\phi\big((Q \cap T) \cup \{g\}\big) \leq \phi(P \setminus T) + [f \notin T]\,\phi_f$*.*
In particular, if $T \supseteq H$ *and* $P \cap T = \emptyset$ *then* $\phi_f \leq \phi(Q \setminus T)$*, and since every* ϕ_i *with* $i \in L$ *is strictly below* ϕ_f *this forces* (1)*:* $|Q \setminus T| \geq 2$*. Symmetrically, if* $T \supseteq H$ *and* $Q \cap T = \emptyset$ *then* $|P \setminus T| \geq 2$*.*

Proof. Cross T with S_1 in Lemma 3: here $S_1 \cap T = (P \cap T) \cup \{f\}$ and $N \setminus (S_1 \cup T) = (Q \setminus T) \cup (\{g\} \setminus T)$, which is (i); (ii) is the mirror. For the last part, $P \cap T = \emptyset$ makes the left side $s_\phi(\{f\}) = \phi_f$, and $g \in T$ deletes the ϕ_g term. If $|Q \setminus T| = 0$ we get $\phi_f \leq 0$, false; if $|Q \setminus T| = 1$ we get $\phi_f \leq \phi_i < \phi_f$, false. □

6 Necessity

By the f/g symmetry, which exchanges (f, P, S_1) with (g, Q, S_2) and carries type (p, q) to type (q, p), it suffices to bound one side.

Theorem 17. *If* $p = 1$ *then* $q \geq 3$*.*

Proof. Write $P = \{a\}$ and suppose $q \leq 2$. By Lemma 7 we may assume $\phi_i > 0$ on L, so no singleton is tight.

Step 1. Lemma 13 at $S = \{a\}$ requires a ϕ-tight T with $|T \cap \{a\}| < \varepsilon_f$, hence $f \in T$ and $a \notin T$.

Step 2: $g \notin T$*.* If $g \in T$ then $T \supseteq H$ and $P \cap T = \emptyset$, so (1) gives $|Q \setminus T| \geq 2$, forcing $q = 2$ and $T \cap Q = \emptyset$, i.e. $T \subseteq H$. But $\{f\}$ is not tight and H is slack. So $T \subseteq Q \cup \{f\}$, and $R := T \cap Q \neq \emptyset$ because $\{f\}$ is not tight.

Step 3. Let d^* be as in Lemma 15; since $p = 1$, $d^*_a = \phi_f > 0$. The normal of $T = R \cup \{f\}$ has $\nu_a = -1$ and $\nu_j = [j \in R]$ for $j \in Q$, so $\nu_T \cdot d^* \geq 0$ reads $\sum_{j \in R} d^*_j \geq d^*_a = \phi_f > 0$; some $j \in R$ has $d^*_j > 0$.

Case $q = 1$*.* Then $R = Q = \{b\}$ and $d^*_b > 0$, so d^* is strictly positive on all of L: a nonzero nonnegative element of $\mathcal{D}$, contradicting Lemma 13.

Case $q = 2$*.* Write $Q = \{b, c\}$ and, relabelling b and c if necessary, take $d^*_b > 0$. If $d^*_c \geq 0$ then again d^* contradicts Lemma 13, so $d^*_c < 0$. Apply Lemma 13 at $S = \{a, b\}$: some ϕ-tight T' has $|T' \cap \{a, b\}| < \varepsilon_f + \varepsilon_g$. Enumerate by $(\varepsilon_f, \varepsilon_g)$.

- $(0,0)$: the right side is 0, impossible.
- $(1,0)$: $a, b \notin T'$ and $f \in T'$, so $T' \subseteq \{c, f\}$ and $T' \neq \{f\}$, whence $T' = \{c, f\}$, with $\nu_a = -1$, $\nu_b = 0$, $\nu_c = 1$. Then $\nu \cdot d^* \geq 0$ gives $d^*_c \geq d^*_a = \phi_f > 0$, against $d^*_c < 0$.
- $(0,1)$: $a, b \notin T'$ and $g \in T'$, so $T' = \{c, g\}$, with $\nu_a = 0$, $\nu_b = -1$, $\nu_c = 0$. Then $\nu \cdot d^* \geq 0$ gives $d^*_b \leq 0$, against $d^*_b > 0$.
- $(1,1)$: $T' \supseteq H$ and T' omits a or b. If $a \notin T'$, Step 2 already excluded it. If $a \in T'$ and $b \notin T'$ then $\nu_a = 0$, $\nu_b = -1$ and $\nu_c = [c \in T'] - 1$. When $c \in T'$, $\nu \cdot d^* \geq 0$ gives $d^*_b \leq 0$, again a contradiction; when $c \notin T'$ we have $T' = H \cup \{a\}$, and crossing that with S_2 gives $T' \cap S_2 = \{g\}$ and $T' \cup S_2 = N$, hence $\phi_g \leq 0$, false.

Every case is contradictory. □

Theorem 18 (necessity)**.** *A boundary partition of type (p, q) requires $p, q \geq 1$ and $p + q \geq 4$.*

Proof. $p, q \geq 1$ was noted after Proposition 6. Theorem 17 gives $p = 1 \Rightarrow q \geq 3$ and, by the f/g symmetry, $q = 1 \Rightarrow p \geq 3$. If $p, q \geq 2$ then $p + q \geq 4$ directly. In all three cases $p + q \geq 4$. □

7 Sufficiency: two closed-form families

Both families are lower envelopes, so exactness and normalization are free by Lemma 5. The leximin verification factors through a single lemma.

Lemma 19 (pinning)**.** *Let v be exact and singleton-normalized on N, let $c > 1$, and put $\phi = \mathbf{1}_L + c\,\mathbf{1}_H$, so $v(N) = m + 2c$. Suppose*

(a) $v(S_1) = \phi(S_1)$ and $v(S_2) = \phi(S_2)$, and these sum to $v(N)$;

(b) the only $x \in \mathrm{Core}(v)$ with $x \geq \mathbf{1}_N$ is ϕ itself.

Then $\phi = \mathrm{Lmin}(v)$, its strata are exactly L at level 1 and then H at level c, the boundary is strict, and S_1, S_2 are ϕ-tight.

Proof. By (a) and Lemma 11, $x(P) + x_f = p + c$ and $x(Q) + x_g = q + c$ throughout the core. Stage 1 maximizes λ such that some core point has all coordinates at least λ; ϕ gives $\lambda \geq 1$, and $\lambda > 1$ would produce a core point strictly above $\mathbf{1}_N$, excluded by (b). So $\mu_1 = 1$, and the feasible set at that level is the single point ϕ, whose coordinates equal 1 exactly on L; hence $B_1 = L$. At stage 2 the players of L are fixed at 1, so $x_f + x_g = (m + 2c) - m = 2c$, the largest common lower bound is c, and it is attained only at $x_f = x_g = c$. Strictness is $1 < c$ and tightness of S_1, S_2 is (a). □

Lemma 19 is Lemma 13 in constructive form: for a two-stratum shape $L \prec H$ the first leximin stage is the only one that is not automatic, so a construction need only control it.

7.1 Family A: both sides large

Let $p, q \geq 2$ and $c = 2$; the common total is $m+4$. Order the coordinates $x_1, \ldots, x_p \mid y_1, \ldots, y_q \mid f, g$, and write

$$\Pi = (2, 2, 1^{p-2}), \qquad \Sigma_a = (0, 2, 1^{q-2}), \qquad \Sigma_b = (0, 4, 1^{q-2}),$$

which sum to $p+2$, q and $q+2$ respectively, every entry of Π being at most 2. The family consists of

A_0	$(\mathbf{1}^p$	$\mid \mathbf{1}^q$	$\mid 2, 2) \;=\; \phi,$
A_1	$(\Pi$	$\mid \Sigma_a$	$\mid 0, 2),$
A_2	$(\Pi$	$\mid \Sigma_b$	$\mid 0, 0),$
A_3^x	$(0$ at x, 2 at one other, 1 elsewhere	$\mid \Sigma_a$	$\mid 2, 2) \quad (x \in P),$
A_4^j	$(\Pi$	$\mid 0$ at y_1, y_j, 1 elsewhere	$\mid 0, 4) \quad (j = 2, \ldots, q).$

Proposition 20. *For all $p, q \geq 2$ the family above realizes a boundary partition of type (p, q), with* $\mathrm{Lmin}(v) = \mathbf{1}_L + 2\,\mathbf{1}_H$ *and* $s_\phi(H) = 4$.

Proof. All entries are nonnegative and every row totals $m + 4$. Each coordinate vanishes in some row — $x \in P$ in A_3^x, y_1 in A_1, y_j in A_4^j, f in A_1, g in A_2 — so Lemma 5 gives an exact singleton-normalized game.

Rigidity. Every row has $g(P)+g_f = p+2$ and $g(Q)+g_g = q+2$: for A_0 these are $p+2$ and $q+2$; for A_1, $(p+2)+0$ and $q+2$; for A_2, $(p+2)+0$ and $(q+2)+0$; for A_3^x, $g(P) = 0+2+(p-2) = p$ and $q+2$; for A_4^j, $(p+2)+0$ and $(q-2)+4$. Hence $v(S_1) = p+2$, $v(S_2) = q+2$, summing to $v(N)$, which is hypothesis (a) of Lemma 19.

Two pins. First, $\max_g(g_{y_1} + g_x) = 2$ for each $x \in P$: row A_0 attains 2, and every other row has $g_{y_1} = 0$ with $g_x \leq 2$. Hence $v(N \setminus \{y_1, x\}) = m + 2$ and every core point obeys $x_{y_1} + x_x \leq 2$. Second, every row has $g_g \geq g_f$ (namely $2 \geq 2$, $2 \geq 0$, $0 \geq 0$, $2 \geq 2$, $4 \geq 0$) with equality at A_0, so $v(\{g\} \cup P) = p + 2$ and every core point obeys $x_g \geq x_f$.

Uniqueness above $\mathbf{1}_N$. Let $x \in \mathrm{Core}(v)$ with $x \geq \mathbf{1}_N$. The first pin and $x_{y_1}, x_x \geq 1$ give $x_{y_1} = 1$ and $x_x = 1$ for all $x \in P$; then rigidity gives $x_f = 2$, the second pin gives $x_g \geq 2$, and rigidity gives $x(Q) = q + 2 - x_g \leq q$. But $x(Q) \geq q$, so $x(Q) = q$, every $x_{y_j} = 1$ and $x_g = 2$. Hence $x = \phi$, which is hypothesis (b).

Slack. $v(H) = \min_g(g_f + g_g) = 0$, attained at A_2, while $\phi(H) = 4$.

Lemma 19 now applies with $c = 2$, and S_1, S_2 are tight, disjoint, covering, with traces $\{f\}$ and $\{g\}$. □

The two hypotheses $p \geq 2$ and $q \geq 2$ are visible in the construction. Π must fit $p + 2$ into p entries each at most 2, which needs $p + 2 \leq 2p$; and that ceiling of 2 is exactly the first pin. At $q = 1$ the row normalizing g would need $g_{y_1} = q + 2 = 3$ against the same ceiling, forcing $g(P) = 0$ against $g(P) = p + 2$. These are the two readings of (1).

7.2 Family B: one side a singleton

Let $p = 1$, $q \geq 3$, and $c = q - 1 \geq 2$; the common total is $3q - 1$. Order the coordinates $a \mid y_1, \ldots, y_q \mid f, g$. Four rows suffice, for every q:

$$\begin{array}{lll} B_0 & (1 & \mid \mathbf{1}^q & \mid c,\ c) \ = \ \phi, \\ B_1 & (0 & \mid q,\ q-1,\ 0^{q-2} & \mid 1+c,\ 0), \\ B_2 & (1+c & \mid 0,\ q-1,\ 0^{q-2} & \mid 0,\ q), \\ B_3 & (1 & \mid 1,\ 0,\ 1^{q-2} & \mid c,\ q). \end{array}$$

Proposition 21. *For all $q \geq 3$ the family above realizes a boundary partition of type $(1, q)$, with* $\mathrm{Lmin}(v) = \mathbf{1}_L + (q-1)\mathbf{1}_H$ *and* $s_\phi(H) = q - 2$.

Proof. Every row totals $3q - 1$, and coordinates vanish: a in B_1, y_1 in B_2, y_2 in B_3, y_k $(k \geq 3)$ in B_1, f in B_2, g in B_1. So Lemma 5 applies.

Rigidity. Every row has $g_a + g_f = 1 + c = q$ and $g(Q) + g_g = q + c = 2q - 1$; this is checked row by row. Hence hypothesis (a) of Lemma 19.

Two pins. Using rigidity, tightness of $\{a, y_1\} \cup H$ is equivalent to $\max_g\, g(Q \setminus \{y_1\}) = q-1$, which holds: the four rows give $q-1$, $q-1$, $q-1$ and $q-2$. So every core point obeys $x(Q \setminus \{y_1\}) \leq q-1$. Likewise tightness of $\{y_2\} \cup H$ is equivalent to $\max_g\,(g_a + g(Q \setminus \{y_2\})) = q$, which holds: the four rows give $1+(q-1)$, $0+q$, $(1+c)+0$ and $1+1+(q-2)$. So every core point obeys $x_a + x(Q \setminus \{y_2\}) \leq q$.

Uniqueness above $\mathbf{1}_N$. Let $x \in \mathrm{Core}(v)$ with $x \geq \mathbf{1}_N$. The first pin bounds a sum of $q-1$ terms each at least 1 by $q-1$, so $x_{y_j} = 1$ for $j \geq 2$. The second bounds a sum of q terms each at least 1

by q, so $x_a = 1$ and $x_{y_j} = 1$ for $j \neq 2$; in particular $x_{y_1} = 1$. Rigidity then gives $x_f = (1+c)-1 = c$ and $x_g = (q+c) - q = c$, so $x = \phi$.

Slack. $v(H) = \min_g(g_f + g_g) = \min\{2c,\ q,\ q,\ c+q\} = q$, while $\phi(H) = 2c = 2q - 2$; the slack is $q - 2$, positive exactly for $q \geq 3$.

Lemma 19 applies with $c = q - 1$. □

At $q = 2$ the same four rows still define an exact game, but two things fail at once: the slack $q - 2$ vanishes, and $c = q - 1 = 1$ makes ϕ constant, so H is not a strict richer suffix. Instantiating them gives rows $(1,1,1,1,1)$, $(0,2,1,2,0)$, $(2,0,1,0,2)$, $(1,1,0,1,2)$ with Lmin $= (1,1,1,1,1)$ and $v(H) = \phi(H) = 2$. This is the constructive shadow of Theorem 17: at $q = 2$ the coalition $\{y_2\} \cup H$ used in the second pin would have to satisfy (1) with $|Q \setminus T| = 1$.

Theorem 22 (sufficiency)**.** *For every $p, q \geq 1$ with $p + q \geq 4$ there is an explicit exact singleton-normalized game carrying a boundary partition of type (p, q).*

Proof. If $p, q \geq 2$ use Family A. If $p = 1$ then $p + q \geq 4$ forces $q \geq 3$, so Family B applies; if $q = 1$ then $p \geq 3$ and the mirror of Family B applies. The three ranges are disjoint and exhaust the hypothesis. □

Theorems 18 and 22 together are Theorem 2.

8 At the threshold

Step 2 of Theorem 17 shows that when $q \leq 2$ no ϕ-tight coalition contains H and omits P. At $q = 3$ the opposite holds.

Theorem 23 (frontier dichotomy)**.** *In every boundary partition of type $(1, 3)$ there is a ϕ-tight T with $H \subseteq T$ and $a \notin T$; necessarily $T = H \cup \{j\}$ for a single $j \in Q$.*

The shape of T is forced by (1), which requires $|Q \setminus T| \geq 2$, together with slackness of H, which excludes $|Q \setminus T| = 3$. Existence is established by an exhaustive finite check over the payoff stratifications, described in Section 10; we have no human proof of it. Together with Step 2 of Theorem 17 the dichotomy reads:

$$q \leq 2: \text{ such a } T \text{ is impossible;} \qquad q = 3: \text{ such a } T \text{ is mandatory.}$$

That is the sense in which the threshold is sharp. The object the obstruction forbids does not become gradually easier to build as q grows; it passes from unavailable to unavoidable at a single step.

Remark 24 (a claim we withdraw)**.** *The crossing budget behind (1) reads $\phi_f \leq \phi(Q \setminus T)$, and in the first type-$(1,3)$ witness we harvested it held with equality. We had recorded that as evidence that the threshold is sharp in a second, quantitative sense. It is not: the budget is in general strict. A search over all stratifications returns 74 stratification–j pairs, each carrying an audited type-$(1,3)$ realization in which it is strict — for instance $\phi = (25/24, 1, 49/48, 17/16, 13/12, 13/12)$ on (a, b, c, d, f, g), with $\max_L \phi = 17/16 < 13/12 = \min_H \phi$, $T = \{c\} \cup H$ tight, and $\phi_f = 13/12 < 33/16 = \phi_b + \phi_d$, a gap of $47/48$. Moreover the same search with the strictness dropped is satisfiable on exactly the same 74 stratification–j pairs, so strictness costs nothing. Theorem 23 is the correct sharpness statement and the equality claim is withdrawn.*

9 The seven-player consequence

We can now settle the zero-payoff question that motivated Definition 1. A census of seven-player games had left one admissible profile undecided, namely $\mathrm{ZP}(2,2)$ with five poorer players; by Proposition 6(v) it descends either to a boundary partition of type $(2,2)$ or to $\mathrm{ZP}(1,2)$ or $\mathrm{ZP}(2,1)$, and those descend in turn to boundary partitions of type $(1,2)$ or $(2,1)$. We had conjectured all three types impossible, which would have refuted the profile. Theorem 2 says otherwise: types $(1,2)$ and $(2,1)$ are indeed impossible, but $(2,2)$ is not.

Example 25 (a type-$(2,2)$ game)**.** *On $N = \{a, \dots, f\}$ take the lower envelope of the five allocations of common total* 8

$$(0,0,1,4,2,1), \quad (0,0,4,4,0,0), \quad (4,0,0,0,2,2), \quad (1,0,2,3,0,2), \quad (0,1,2,3,1,1).$$

Then v is exact and singleton-normalized, $\mathrm{Lmin}(v) = (1,1,2,2,1,1)$ *with strata $\{a,b,e,f\} \prec \{c,d\}$, so $H = \{c,d\}$ is a strict richer suffix with $m = 4$; $v(cd) = 0$, so the boundary is slack with $s_\phi(H) = 4$; the proper tight coalitions are ad, abd, ade, $acde$, adf, $acdf$, cef, def, $acdef$; and $S_1 = cef$, $S_2 = abd$ are tight, disjoint, cover N, and meet H in $\{c\}$ and $\{d\}$. Their poorer parts are $\{e,f\}$ and $\{a,b\}$, so the type is $(2,2)$.*

Corollary 26. *There is a seven-player exact singleton-normalized game carrying a* $\mathrm{ZP}(2,2)$ *configuration with five poorer players.*

Proof. Adjoin a null player g to Example 25 — concretely, append a seventh coordinate 0 to each of the five rows. By Proposition 6(iv) the result is exact and singleton-normalized with $\mathrm{Lmin} = (1,1,2,2,1,1,0)$, strata $\{g\} \prec \{a,b,e,f\} \prec \{c,d\}$, the same boundary $H = \{c,d\}$ with slack 4, and tight sets $cefg$ and $abdg$ meeting in $\{g\}$ and covering N; the poorer parts are $\{e,f\}$ and $\{a,b\}$. □

Family A of Section 7 gives an alternative, uniform witness at $(2,2)$, with $\mathrm{Lmin} = \mathbf{1}_L + 2\mathbf{1}_H$; Example 25 is retained because it is the game that refuted the conjecture and because its tight family is visibly different, which is a useful reminder that the families of Section 7 are constructions and not normal forms.

Remark 27 (why the census missed it)**.** *Type $(2,2)$ was absent from a census of some* 20,000 *sampled slack boundaries with $m = 4$; a later targeted harvest found it once in* 15,967 *games, against six of type $(1,3)$ and seven of type $(3,1)$. It is rare, not absent. The general lesson — that "configuration X never occurs" is a statement about a sampler until it is a statement about set systems — is recorded for the same reason in (Flanagan, 2026).*

10 What is proved, what is checked, what is verified

status	statement
proved	Theorems 8 and 9; Lemmas 4, 11, 12, 13, 15, 16, 7, 19; Proposition 6; Theorems 17 and 18; Propositions 20 and 21 and Theorem 22; hence Theorem 2. None of these consults a machine.
finite check	Theorem 23: unsatisfiable over all 60 payoff stratifications of type $(1, 3)$, using the encoding of Theorems 8 and 9, with a satisfiable control confirming the encoding is not vacuous.
verified	Example 25 and Corollary 26, by exact rational recomputation of the envelope, of exactness at definition level, of Lmin, of the tight family and of the configuration; the 74 strict-budget realizations (one per stratification–j pair) of Remark 24, each audited the same way; and the two families of Section 7 at all 18 instances with $n \leq 9$.

An earlier route to Theorem 18 was a finite check — 48 unsatisfiable instances at $m = 3$ over the encoding of Section 4, cross-checked by an independent enumeration of tight families — and it is retained in the repository as a control on the hand proof rather than as a dependency. Likewise the padding construction that first established sufficiency (adjoin a null player and place it on either side of the partition) remains true but is no longer load-bearing, having been superseded by Section 7.

11 Scope

Four limitations are worth stating plainly.

First, the characterization is specific to $|H| = 2$. For $|H| \geq 3$ a "boundary partition" would have to be defined with more than two blocks, and Lemma 11 would still apply, but neither the necessity argument nor the families have been examined in that setting.

Second, Theorem 23 is a finite check, not a theorem with a human proof. It is not used anywhere else in this note; nothing in Theorem 2 depends on it.

Third, the families of Section 7 carry no minimality claim. Family A uses $m+2$ rows and Family B uses four, but we prove no lower bound on the number of generators, and Family A's uniform shape — payoffs 1 and 2, slack 4 independent of p and q — is a property of that construction rather than of the configuration. Example 25 and the configurations of Remark 24 look nothing like it.

Fourth, we characterize which types occur, not how often or in what proportion of the exact cone. The census figures quoted in Section 9 are descriptive of a sampler.

Reproducibility note

Theorem 2 and everything it rests on are proved by hand; the families of Section 7 are finite verifications of row arithmetic, uniform in (p, q). The one proof dependency that is a machine result is Theorem 23, and it is used nowhere else. Exact rational programs archived in `../verifiers/part` reconstruct, independently of the text: the game of Example 25 and its seven-player suspension, including exactness at definition level by linear programming over the core rather than by vertex enumeration; the two families at every instance with $n \leq 9$; the finite check behind Theorem 23

together with its control; the withdrawn-equality search of Remark 24; and the superseded $m = 3$ finite check with its independent cross-check. All arithmetic is exact; every solver call is asserted satisfiable or unsatisfiable, and no call returned an indeterminate result. No core-vertex enumeration is performed at any n.

Draft note on references. This list contains only the works actually cited above. Every entry is carried over verbatim from the companion paper's bibliography and still requires the editorial verification pass; no reference here was reconstructed from memory. See the prior-art paragraph in Section 1 for the citations this note still needs.

References

Flanagan M (2026) Balanced transport at the egalitarian core: a complete boundary phase diagram. Companion manuscript.

Lohmann E, Borm P, Herings PJJ (2012) Minimal exact balancedness. Mathematical Social Sciences 64:127–135.

Studený M, Kratochvíl V (2022) Facets of the cone of exact games. Mathematical Methods of Operations Research 95:35–80.